

LECTURE #9

Econometrics I

QUALITATIVE ANALYSIS & LINEAR PROBABILITY MODEL

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In the previous lecture #8

- ▶ We summarized **four important variable selection criteria**:
 1. theory, 2. OVB reduction, 3. $\bar{R}^2$, 4. t/F test.
- ▶ For **predictions**, we derived the confidence/prediction intervals:
 - ▶ mean/average prediction $\mathbb{E}(y|x_{n+1})$ vs.
 - ▶ prediction for a specific unit: $se(\hat{e}^0) = \sqrt{\hat{\sigma}^2 + \text{Var}(\hat{y}^0)}$.
- ▶ We introduced regression with **qualitative information**:
 - ▶ intercept dummy:
$$D_i = 1 : y_i = (\beta_0 + \beta_k) + \beta_1 x_{i,1} + \dots + \beta_{k-1} x_{i,k-1} + u_i,$$
 - ▶ dummy variable trap (base group),
 - ▶ multiple categories:
$$y = \beta_0 + \beta_1 x + \beta_2 D_1 + \beta_3 D_2 + \beta_4 D_3 + u,$$
 - ▶ ordinal information: $Q = 0, 1, 2, 3$.
- ▶ Readings for lecture #9:
 - ▶ Chapter 7: 7.4–7.7.

Home assignment #2

- ▶ Assigned on Wed/Thu after the seminars via SIS.
- ▶ Teams of two, one report.
- ▶ [‘Matching spreadsheet’](#).
- ▶ Delivered electronically in the .pdf format [one file, 20 MB max, .R or other formats can be delivered together in .zip] via the **‘Study group roster’** app (Lecture JEB109) in SIS.
- ▶ Deadline: Monday, May 4, 2026, 23:59:59.
- ▶ ‘Academic integrity’; solo $\Rightarrow$ 0.
- ▶ IES guide: [AI tools when studying at IES](#).

Outline

More on binary independent variables

Interactions involving dummy variables

Testing for differences between groups: Chow test

Binary dependent variable: Linear probability model (LPM)

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Interactions among dummy variables

- ▶ Similarly, as for quantitative explanatory variables, we can create interactions between/across dummy variables and between/across dummy and quantitative variables.
- ▶ The former is practically another way of forming multi-criterial dummies as these two approaches are effectively equivalent:
 - ▶ multi-criterial dummy variables from lecture #8:

$$income = \gamma_0 + \gamma_1 UC + \gamma_2 EF + \gamma_3 UF + v$$

are more suitable for testing differences between groups,

- ▶ interacting dummies are more suitable for testing a general statement of significant interactions, e.g., whether employment income differentials depend on citizenships (discrimination),
- ▶ care must be taken with setting and understanding the base group correctly,
- ▶ when including interaction among dummies, the separate ones are usually part of the model as well.

Interactions among dummy variables: Example

- ▶ Back to our example, we can have the following specification:

$$income = \beta_0 + \beta_1 E + \beta_2 C + \beta_3 E \cdot C + v.$$

- ▶ Figuring out the base group:
 - ▶ when dummy variables and the interaction products are zeros,
 - ▶ in this case, we need $E = 0$ & $C = 0$,
 - ⇒ our base group is an unemployed foreigner.
- ▶ To test against the other groups, we need to pick carefully:

employed citizens	$E = 1$ & $C = 1$	test $\theta \equiv \beta_1 + \beta_2 + \beta_3 = 0$,
unemployed citizens	$E = 0$ & $C = 1$	only test $\beta_2 = 0$,
employed foreigners	$E = 1$ & $C = 0$	only test $\beta_1 = 0$.
- ▶ Apparently, this can get a bit complicated.

Allowing for different slopes: Slope dummy

- ▶ We already know that a separate intercept dummy allows for different intercepts across multiple categories.
- ▶ Interaction term between a dummy and a quantitative explanatory variable functionally changes the model to allow for different slopes for specific groups.
- ▶ Consider the following model:

$$y = \beta_0 + \beta_1 x + \beta_2 D + \beta_3 Dx + u.$$

- ▶ Can this model be estimated? Is there a problem with the dummy variable trap?
- ▶ This, in practice, gives us two models for two subpopulations:

$$D_i = 0 \quad : \quad y_i = \beta_0 + \beta_1 x_i + u_i,$$

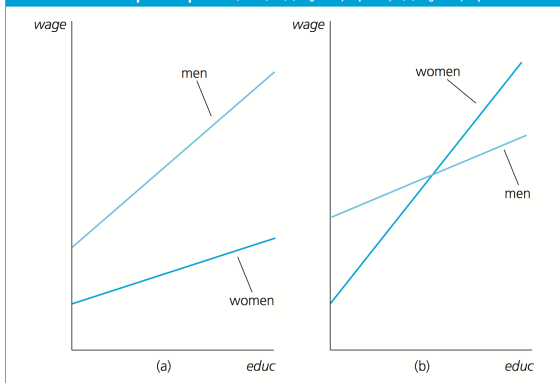
$$D_i = 1 \quad : \quad y_i = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)x_i + u_i.$$

- ▶ This specification allows for both the individual and joint significance testing (be aware of high sample correlation between D and Dx).

Illustration: Different intercepts vs. different slopes

$$\begin{aligned} \text{wage} &= \beta_0 + \delta_0 \text{female} + \beta_1 \text{educ} + \delta_1 \text{female} \cdot \text{educ} + u \\ &= (\beta_0 + \delta_0 \text{female}) + (\beta_1 + \delta_1 \text{female}) \text{educ} + u \end{aligned}$$

FIGURE 7.2 Graphs of equation (7.16): (a) $\delta_0 < 0$, $\delta_1 < 0$; (b) $\delta_0 < 0$, $\delta_1 > 0$.



Source: Wooldridge (2012)

Illustration in R: Two subpopulations

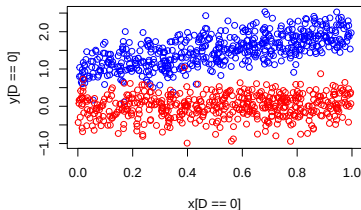
$$y = (\beta_0 + \delta_0 D) + (\beta_1 + \delta_1 D)x + u$$

$$\text{setup} : \beta_0 = 1, \delta_0 = -1, \beta_1 = 1, \delta_1 = -1$$

$$D_i = 0 : y_i = \beta_0 + \beta_1 x_i + u_i$$

$$D_i = 1 : y_i = u_i$$

(a) $\sigma^2 = 0.1$



(b) $\sigma^2 = 0.5$

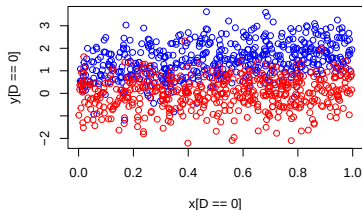


Illustration in R: Two subpopulations

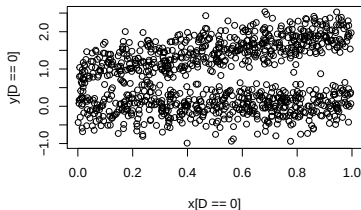
$$y = (\beta_0 + \delta_0 D) + (\beta_1 + \delta_1 D)x + u$$

$$\text{setup} : \beta_0 = 1, \delta_0 = -1, \beta_1 = 1, \delta_1 = -1$$

$$D_i = 0 : y_i = \beta_0 + \beta_1 x_i + u_i$$

$$D_i = 1 : y_i = u_i$$

(a) $\sigma^2 = 0.1$



(b) $\sigma^2 = 0.5$

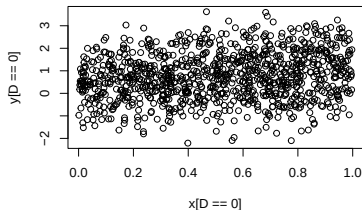


Illustration in R: Two subpopulations: (a) $\sigma^2 = 0.1$

setup : $\beta_0 = 1, \delta_0 = -1, \beta_1 = 1, \delta_1 = -1$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.97345	0.02921	33.33	<2e-16 ***
D	-1.03492	0.04163	-24.86	<2e-16 ***
x	1.01965	0.04978	20.48	<2e-16 ***
I(D * x)	-0.90469	0.07086	-12.77	<2e-16 ***

Multiple R-squared: 0.857, Adjusted R-squared: 0.8566

vs.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.47103	0.05393	8.734	< 2e-16 ***
x	0.57626	0.09181	6.276	5.16e-10 ***

Multiple R-squared: 0.03797, Adjusted R-squared: 0.03701

Illustration in R: Two subpopulations: (a) $\sigma^2 = 0.5$

setup : $\beta_0 = 1, \delta_0 = -1, \beta_1 = 1, \delta_1 = -1$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.94063	0.06532	14.401	< 2e-16	***
D	-1.07809	0.09308	-11.583	< 2e-16	***
x	1.04394	0.11131	9.379	< 2e-16	***
I(D * x)	-0.78687	0.15845	-4.966	8.04e-07	***

Multiple R-squared: 0.5416, Adjusted R-squared: 0.5403

vs.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.41680	0.06756	6.169	9.96e-10	***
x	0.65866	0.11502	5.726	1.36e-08	***

Multiple R-squared: 0.03181, Adjusted R-squared: 0.03084

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Testing for differences between groups

- ▶ Testing among various dummy variable parameters allows us to uncover differences between groups.
- ▶ **Chow test** is frequently used to test the stability/equality of the parameters of the underlying population model for different groups.
- ▶ Assume the population model with $k = 1$: $y = \beta_0 + \beta_1 x + u$.
- ▶ Let us assume two subpopulations and define a dummy

$$D_i = \begin{cases} 0 & \text{if the } i\text{-th observation is from the first subpopulation,} \\ 1 & \text{if the } i\text{-th observation is from the second subpopulation.} \end{cases}$$

- ▶ Extend the population model

$$\begin{aligned} y_i &= \beta_0 + \delta_0 D_i + \beta_1 x_i + \delta_1 D_i x_i + u_i \\ &= (\beta_0 + \delta_0 D_i) + (\beta_1 + \delta_1 D_i) x_i + u_i. \end{aligned} \quad (1)$$

- ▶ Using an F test, we can specify

$$H_0 : \delta_0 = 0 \text{ and } \delta_1 = 0 \quad \text{vs.} \quad H_1 : \delta_0 \neq 0 \text{ or } \delta_1 \neq 0.$$

Chow test

- ▶ For a standard F test, $D_i = 1$ defines the unrestricted model (1) while $D_i = 0$ defines the restricted model:

$$y_i = \beta_0 + \beta_1 x_i + u_i. \quad (2)$$

- ▶ If we estimate the restricted/**population model** (2) **separately** over the two subpopulations/**subsamples** and obtain the respective SSR_1 and SSR_2 , it can be shown that $SSR_1 + SSR_2 = SSR_U$ from (1).
- ▶ SSR_R based on (1) is then simply the residual sum of squares from (2) estimated using the **pooled** dataset, which we label SSR_P .
- ▶ F statistic, or the **Chow statistic**, is then defined as

$$F = \frac{SSR_P - (SSR_1 + SSR_2)}{SSR_1 + SSR_2} \frac{n - 2(k + 1)}{k + 1},$$

compare to $F = \frac{SSR_R - SSR_U}{SSR_U} \frac{n - k - 1}{q}.$

- ▶ Under the null hypothesis, i.e., that the population model does not differ for the subpopulations, $F \sim F_{k+1, n-2(k+1)}.$

▶ Gregory Chow

Chow test* (alternative, less strict)

- ▶ As for a standard F test, we assume homoskedasticity and normality of the error term for an exact F distribution under the null, but the normality assumption can be dropped for large samples, and the statistic is then approximately F -distributed.
- ▶ Sometimes, the classical Chow test can be too strict as it does not allow any parameter to differ for subgroups, including the intercept.
- ▶ Alternative test specification and its testing statistic is written as

$$F = \frac{SSR_P^* - (SSR_1 + SSR_2)}{SSR_1 + SSR_2} \frac{n - 2(k + 1)}{k},$$

where SSR_P^* is the residual sum of squares of the restricted/population model (3), which, however, includes a dummy variable for the **intercept shift**, i.e., we consider **one less restriction**:

$$y_i = (\beta_0 + \delta_0 D_i) + \beta_1 x_i + u_i. \quad (3)$$

- ▶ This also changes the limiting distribution to $F \sim F_{k, n-2(k+1)}$.

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Binary dependent variable

- ▶ Until now, we have only used qualitative variables as the independent ones in a model.
- ▶ However, also various dependent variables of interest have a **qualitative** or discrete nature (this semester, we only stick to the binary ones).
- ▶ Population model remains the same as for a quantitative dependent variable:

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u,$$

but with y being binary, i.e., only 1 or 0.

Expected value of y

- ▶ As y is either 1 or 0, the interpretation of β s changes.
- ▶ Under MLR.1–4, the OLS estimator is still unbiased and consistent, and we thus still have

$$\mathbb{E}(y|X) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k.$$

- ▶ Importantly, y is a discrete random variable with a Bernoulli distribution

$$y = \begin{cases} 1 & \text{with probability } P, \\ 0 & \text{with probability } 1 - P. \end{cases}$$

- ▶ We can thus find the expected value

$$\boxed{\mathbb{E}(y|X)} = 1 \cdot P(y = 1|X) + 0 \cdot (1 - P(y = 1|X)) = \boxed{P(y = 1|X)}.$$

- ▶ I.e., the expected value of the dependent variable y (given X) is the probability of y being 1 (given X).

Interpretation of β s in the LPM framework

- ▶ $P(y = 1|X)$ is usually referred to as the **response probability**.
- ▶ Both $P(y = 1|X)$ and $P(y = 0|X) = 1 - P(y = 1|X)$ are linear in parameters β_j and usually also in all x_j , so the model is called the **linear probability model (LPM)**

$$p(X) \equiv P(y = 1|X) = \mathbb{E}(y|X) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k.$$

- ▶ Previous derivations allow for the interpretation of β s as

$$\Delta p(X) = \Delta P(y = 1|X) = \beta_j \Delta x_j,$$

i.e., the change in the **probability of ‘success’** (probability of y being 1) when x_j changes by one small unit.

- ▶ $\hat{y}$ is the predicted probability of ‘success’.
- ▶ Dummies can be added as independent variables as well.

Shortcomings of the LPM

1. Estimated/predicted probability is **not bounded by 0 and 1**.
 - ▶ this can be partially solved either by (a) censoring to $\langle 0, 1 \rangle$ or by (b) thresholding, i.e., setting a value of $\hat{y}$ separating 0 and 1,
 - ▶ both potentially problematic: (a) too many 0 and 1, i.e., exact predicted probabilities 0% and 100%; (b) how to set the threshold?
2. Usually **constant marginal effect** of Δx_j (often unrealistic).
3. Error term is inherently **heteroskedastic**:
 - ▶ MLR.5 violated, but OLS remains unbiased and consistent,
 - ▶ heteroskedasticity needs to be dealt with (lecture #10) as it is crucial for justifying the usual t and F statistics even in large samples.

$$\begin{aligned}\boxed{\text{Var}(u|X)} &= \text{Var}(y|X) = \mathbb{E}(y^2|X) - (\mathbb{E}(y|X))^2 = \\ &= 1^2 \cdot p(X) + 0^2 \cdot (1 - p(X)) - (p(X))^2 = \boxed{p(X)(1 - p(X))}.\end{aligned}$$

4. Error term is **not normally distributed**.
 - ▶ because y takes on only two values, u also takes on only two possible values for given X .

Potential advantages of the LPM

Nonetheless, the LPM is still useful and often applied in economics, because:

1. Simple to estimate (OLS still unbiased and consistent).
2. Intuitive, straightforward, and generally OK linear interpretation.
3. Heteroskedasticity and nonnormality can be dealt with.
4. It usually works well for values of x_j relatively close to the sample averages.

However, be aware: R^2 and $\bar{R}^2$ no longer good measures of the goodness-of-fit for the LPM!

Seminars and the next lecture

- ▶ Seminars:
 - ▶ intercept and slope dummies,
 - ▶ multiple categories and interactions with dummies,
 - ▶ model construction,
 - ▶ practicing Chow test in R,
 - ▶ LPM in practice in R.
- ▶ Next lecture #10:
 - ▶ heteroskedasticity: consequences for OLS,
 - ▶ heteroskedasticity-robust inference,
 - ▶ testing for heteroskedasticity,
 - ▶ weighted least squares (WLS) estimator,
 - ▶ LPM revisited,
 - ▶ (F)GLS estimator.
- ▶ Readings for lecture #10:
 - ▶ Chapter 8 (**8.4, sections ‘What If...’ & ‘Prediction and...’, and 8.5 mandatory after lecture**).

Historical spotlight: Gregory Chow (1930)

A Chinese-American economist Gregory Chi-Chong Chow made a simple question famous: “Do we have one universal relationship, or did something change?”

After MIT and Cornell, he worked at the IBM Thomas J. Watson Research Center while also advising the government of Taiwan. Later, he joined Princeton University. Working as an econometrician at Princeton and Xiamen University and later advising on China's economic reforms, he turned that question into the Chow test, a clean check for a structural break (that is, the stability of regression coefficients across different data samples). You fit one regression to the whole sample, then fit two regressions to two periods or groups, and compare how much the residuals shrink. If the two-fit version leads to a significant reduction of RSS, the F -statistic indicates that the coefficients differ and the relationship has shifted.

Think of a policy change, a new technology, or an oil shock. Before the break, prices and output may move in one direction; after the break, the slope or intercept may change. The Chow test turns your opinion ‘it looks different’ into a value backed by statistical principles you can argue with.

Why it matters for econometrics: models age or might differ for subpopulations. The Chow test provides a quick way to check stability, select sensible breakpoints, and determine whether to pool data or estimate separately.



Source: AEAweb.org